

1.5 Review Questions

Short Answer Questions

1. What is Intelligence?
2. What is Artificial Intelligence?
3. Mention any two definitions of Artificial Intelligence.
4. List the types of Artificial Intelligence.
5. Give any two examples of Artificial Intelligence applications.
6. Write any two technologies used in Artificial Intelligence.

7. Mention any two advantages of Artificial Intelligence.
8. Mention any two disadvantages of Artificial Intelligence.
9. What is the main goal of Artificial Intelligence?
10. Name any two fields where Artificial Intelligence is widely used.
11. What is an Intelligent Agent?
12. What is the role of an environment in AI?
13. Mention any two types of agents based on behaviour.
14. Define Rationality in AI.
15. Mention any two factors that affect the rationality of an agent.
16. Differentiate between Rational Agent and Irrational Agent (one point).
17. What is an environment in Artificial Intelligence?
18. What does PEAS stand for?
19. Mention any two properties of an environment.
20. What is meant by the structure of an intelligent agent?
21. What is an Agent Program?
22. Differentiate between Agent Function and Agent Program (one point).
23. What is the need for designing intelligent agents?
24. What is a Simple Reflex Agent?
25. Define Model-Based Reflex Agent.
26. What is a Goal-Based Agent?
27. What is a Utility-Based Agent?
28. Define a Learning Agent.
29. Mention any two features of Learning Agents.
30. Write the full form of PEAS with its meaning.
31. What is Knowledge?
32. Mention two types of Knowledge in AI.
33. What is a Knowledge-Based Agent?
34. Write any two key characteristics of Knowledge-Based Agents.
35. Mention the main components of a Knowledge-Based Agent.
36. What is the purpose of a Knowledge Base?
37. What are the functions supporting the Knowledge Representation Language?
38. What is the role of an Inference Engine?
39. Write any two levels of a Knowledge-Based Agent.
40. What is Wumpus World in AI?
41. Mention two percepts used in Wumpus World.

42. What does "P" represent in the PEAS description of Wumpus World?
43. Mention any two properties of the Wumpus World.
44. What is a Problem-Solving Agent?
45. Mention two main functions of a Problem-Solving Agent.
46. What kind of problems are solved by a Problem-Solving Agent?
47. Write any two characteristics of a Problem-Solving Agent.
48. Mention any two steps performed by a Problem-Solving Agent.
49. What is a State Space in AI problem solving?
50. Define Initial State and Goal State.
51. What is a Path Cost?
52. What is a Toy Problem?
53. Name any two Toy Problems in AI.
54. Mention any two Real-World Problems solved by AI.
55. What is the Traveling Salesperson Problem?
56. What is the Water Jug Problem?
57. What is the main objective of the Robot Navigation Problem?

Long Answer Questions

1. Explain the meaning of Intelligence and Artificial Intelligence with examples.
2. Write and explain any four definitions of Artificial Intelligence.
3. Describe the different types of Artificial Intelligence.
4. List and explain the top technologies used in Artificial Intelligence.
5. Discuss the major advantages and disadvantages of Artificial Intelligence.
6. Explain the importance of Artificial Intelligence in modern life.
7. Define Intelligent Agents. Explain their relationship with environments.
8. Describe the classification of agents based on their behaviour or characteristics.
9. Explain the concept of Rationality and list the factors that affect it.
10. Differentiate between Rational and Irrational Agents with examples.
1. Explain the PEAS Framework used to specify the task environment.
2. Describe the properties of an environment in Artificial Intelligence.
3. Explain the structure of an Intelligent Agent with a neat diagram.
4. Discuss the difference between Agent Function and Agent Program.
5. Explain the working and features of Simple Reflex Agents.
6. Describe Model-Based Reflex Agents with their working principle.
7. Explain Goal-Based Agents and their importance in AI.

18. Explain the working and features of Utility-Based Agents.
19. Describe the components and working of a Learning Agent.
20. Explain how the components of Agent Programs work together.
21. Define Knowledge and explain the different types of Knowledge in AI.
22. What are Knowledge-Based Agents? Explain their key characteristics.
23. Describe the components of a Knowledge-Based Agent.
24. Explain step by step how a Knowledge-Based Agent works.
25. Write about the main actions performed by a Knowledge-Based Agent.
26. Describe the functions supporting the Knowledge Representation Language.
27. Explain the various levels of a Knowledge-Based Agent.
28. Discuss different approaches to designing a Knowledge-Based Agent.
29. Explain the Wumpus World as an example of a Knowledge-Based Agent.
30. Describe the percepts, clues, and PEAS representation of the Wumpus World.
31. Write about the properties and exploration process of the Wumpus World.
32. Explain the Knowledge Base used in the Wumpus World example.
33. Define Problem-Solving Agents and explain their importance in AI.
34. Discuss the types of problems solved by Problem-Solving Agents.
35. Describe the main functions and characteristics of a Problem-Solving Agent.
36. Explain the steps performed by a Problem-Solving Agent in detail.
37. Write short notes on the following terms:
 - a) State Space
 - b) Goal State
 - c) Path Cost
38. Explain the 8-Puzzle Problem with suitable illustrations.
39. Describe the 8-Queens Problem and how AI approaches are used to solve it.
40. Explain the Water Jug Problem and how it is represented as a state-space problem.
41. Discuss the Traveling Salesperson Problem and how AI can find an optimal path.
42. Explain the Robot Navigation Problem and how a problem-solving agent helps.

